UI/UX AUDIT REPORT

Issac Healthcare AI Platform

A comprehensive UI/UX audit covering information architecture, visual design systems, accessibility compliance, interaction design, and a strategic roadmap to achieve a world-class medical learning experience.

CONDUCTED BY

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DATE

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PLATFORM

Issac — AI Medical Learning & Intelligence

METHODOLOGY

Nielsen's Heuristics · WCAG 2.1 AA · Competitive Analysis

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Structured findings, analysis, and actionable roadmap

SECTION 01

Executive Summary

The **Issac Healthcare AI Platform** is a premium, glassmorphic medical learning application built for clinical scholars. It features rich visual aesthetics — custom radial glows, neumorphic cards, modern typography via Outfit & Plus Jakarta Sans — and high-density clinical data.

While the platform succeeds in creating a stunning first impression and offers a wide breadth of clinical content, a deep UI/UX audit reveals **critical usability defects**, **broken interaction loops**, **visual design regressions across themes**, and **severe accessibility violations** that prevent it from meeting production-quality standards.

> **Audit Methodology**
> This audit evaluates the platform against **Nielsen's 10 Usability Heuristics**, **WCAG 2.1 AA Guidelines**, and modern interaction design standards. Findings are sourced from a full code review of the HTML (1,657 lines), CSS (2,742 lines), and JavaScript (4,253 lines) codebase, supplemented by headless browser screenshot captures.

Key Findings at a Glance

SEVERITY	COUNT	HIGHLIGHTS
CRITICAL	5	Broken case simulator learning loop, keyboard navigation deficit, invisible hover text, useless selector bug, muted text contrast failures
HIGH	6	Simulator layout imbalance, legacy dead-end panel, missing bookmark feedback, abrupt search redirects, no touch events on mobile, inline style design debt
MEDIUM	4	Sidebar cognitive overload, artificial loading delay, native alert dialogs, splash overflow on small screens
LOW	2	Missing avatar alt text, comparison DB key naming conventions

SECTION 02

UX Scorecard by Workflow

Onboarding & Splash	8/10
Dashboard UX	7/10
Topic Learning	6.5/10
Search & Discovery	5/10
Clinical Simulator	3/10
Concept Comparison	2/10
Accessibility (a11y)	2/10
Mobile & Responsive	4.5/10

SECTION 03

Information Architecture & Navigation System

MEDIUM Sidebar Navigation Cognitive Overload

The left sidebar presents **14 global navigation items** across three groups (Workspace, Medical AI Engine, Studies & Learning). Miller's Law states that the average person can only keep 7 (± 2) items in working memory.

Items like *Diagram Studio*, *Medical Flowcharts*, *Concept Compare*, and *Smart Notes Gen* are contextual widgets — not global navigation destinations. Their presence at the top level inflates cognitive load and slows wayfinding.

RECOMMENDATION

Consolidate 14 items into **5 primary navigation hubs**: Dashboard, Learning Engine, AI Workspace, Simulation Labs, and Creator Studio. Nest contextual widgets within their parent hubs.

HIGH Legacy Dead-End Panel: Concept Maps

The Dashboard Quick Actions include a "Concept Maps" button that navigates to a **legacy wrapper panel** (`panel-concept-maps`). This panel renders a bare container whose only interactive element is a button saying "Open Knowledge Engine".

This creates a **dead-end workflow**: the user clicks "Concept Maps", arrives at an empty screen, and is forced to click again to reach the actual Knowledge Engine. Meanwhile, "Knowledge Engine" is already available in the sidebar.

Dashboard Quick Actions → **Concept Maps** Dead-End Panel → **Knowledge Engine** Active Workspace

RECOMMENDATION

Remove the legacy `panel-concept-maps` entirely. Update the Quick Action button to navigate directly to the Knowledge Engine panel.

SECTION 04

Screen-by-Screen Usability Audit

4.1 Vision Splash Screen (Landing Page)

The landing page is the strongest screen in the platform. Premium dark-mode radial gradients, animated glow pulses, and the SVG neural-network brand logo set a high-end tone. The CTA button placement above the features grid is excellent.

MEDIUM **Splash Overflow on Small Viewports**

The splash screen uses a fixed `100vh` height. On smaller laptop screens (11–13"), the features grid overflows and gets clipped because vertical margins are rigid pixel values rather than responsive units.

4.2 Clinical Case Study Simulator

CRITICAL **Broken Educational & Learning Loop**

The patient file stack (Patient Summary, Clinical Findings, Differentials, Investigations, Management, AI Insights) is **fully visible and readable** from the very start of a clinical simulation.

The simulator questions ask the user to diagnose the patient, select the correct test, and establish a treatment plan. However, the exact answers are written in the cards directly below the chat box (e.g., "ECG reveals 3mm ST-segment elevation", "Primary PCI of LAD").

Root Cause

The `updateCardLockStates()` method in `app.js` (line 2958) **unconditionally removes** the `.locked` class from all patient file cards, disabling the blur/lock overlay progression that was originally designed to mask answers.

IMPACT

This completely destroys the diagnostic challenge and recall value. The user has zero incentive to think critically, as they are presented with a cheat sheet on the same screen. This is the single most impactful UX defect in the application.

HIGH

Severe Layout Imbalance

The simulator uses a two-column grid (`1.2fr 1.3fr`). The left column stacks the Chat Simulator, XP stats, and **all 6 patient file cards**, making it ~2000px high. The right column has only 3 short static reference cards (~500px high), creating a massive empty space.

```
┌──────────────────────────────────────────────────────────────┐
│                    Filter Bar (Search + Filters)              │
├───────────────────────────┬──────────────────────────────────┤
│                           │                                  │
│  [Chat Simulator]         │  [Clinical Pearls]               │
│                           │  [Red Flags Tips]                │
│  [Patient Summary]        │  [Related Cases]                 │
│  [Clinical Findings]      │                                  │
│  [Differentials]          │  ▓▓ MASSIVE EMPTY SPACE ▓▓        │
│  [Investigations]         │                                  │
│  [Management]             │                                  │
│  [AI Insights]            │                                  │
│                           │                                  │
└───────────────────────────┴──────────────────────────────────┘
```

RECOMMENDATION

Place the Chat Simulator on the left and the Patient Case File cards on the right. Move static reference cards into collapsible tabs or a secondary drawer to balance column heights.

4.3 Concept Comparison Screen

CRITICAL **The Useless Duplicate Selector Bug**

The screen displays two dropdown selectors (`#compare-select-1` and `#compare-select-2`) with the label "Compare Pairs: [Dropdown 1] with [Dropdown 2]". Both dropdowns are populated with pre-made *pairs* (e.g., "STEMI vs NSTEMI Infarction").

When the user clicks "Compare", the JavaScript code **only reads the value from the first selector** and completely ignores the second one. Selecting a concept in the second dropdown has zero effect.

> **Design Incoherence**
>
> The layout implies the user is selecting two independent concepts to compare (Concept A *with* Concept B), but the actual data model uses pre-paired entries. The second selector is functionally dead code.

RECOMMENDATION

Remove the second selector entirely. Relabel the first selector to "Select Comparison Pair:" to match the actual data model.

4.4 Search & AI Assistant Workspace

HIGH **Forced Loading Delay & Abrupt Auto-Redirect**

Every search query initiates a progress bar animation that takes **3.8 seconds** before displaying results. While acceptable for initial onboarding, this artificial delay becomes a workflow bottleneck for power users.

When the animation completes, the application abruptly switches from the AI Workspace panel to `panel-global-search` without user consent. This sudden screen transition violates the heuristic of **User Control and Freedom**.

SECTION 05

Visual Design System & Consistency

CRITICAL

Light Mode Hover Text Invisibility

The `.btn-outline` class sets invisible borders in light mode (`rgba(255, 255, 255, 0.1)` on white). On hover, the text color changes to white:

```
.btn-outline:hover {
  background: rgba(255, 255, 255, 0.02);
  color: #fff;  /* White text on white background! */
  border-color: rgba(255, 255, 255, 0.2);
}
```

In light mode, hovering over buttons like "Explore All", "Generate Notes", or "← Back to Topic" makes their text **completely invisible**.

ROOT CAUSE

The stylesheet was authored exclusively for a dark theme and lacks light-mode class overrides for interactive button hover states. No `body:not(.dark-mode)` or `body.dark-mode` scope exists for `.btn-outline` .

HIGH

Missing Bookmark Visual Feedback

Clicking the bookmark button in Topic Learning correctly adds/removes the `.active` class and updates the internal array. However, **no CSS rules** are defined for `.bookmark-btn.active` in `styles.css` .

The bookmark button looks identical whether the topic is bookmarked or not. This violates Nielsen's heuristic of **Visibility of System Status** — the user cannot verify if their action succeeded without navigating to the Library panel.

HIGH

Inline Style Design Debt

Design classes like `.badge-green` and `.badge-blue` are routinely overridden by inline styles in the HTML to force indigo brand colors (e.g., `style="background: var(--primary-glow); color: var(--primary);"`).

This fragments the design system, inflates HTML file size (106KB), and makes dark/light mode synchronization prone to regressions since inline styles cannot be overridden by theme classes.

SECTION 06

Accessibility & WCAG Compliance

Overall Assessment: Non-Compliant

The application fails multiple WCAG 2.1 Level A and AA success criteria. Keyboard-only users are completely locked out of the primary navigation and all interactive diagrams.

CRITICAL

WCAG 2.1.1: Keyboard Navigation Deficit

The global sidebar navigation links are implemented as `<li>` elements **without** `tabindex` attributes and without keyboard event handlers. Only mouse click listeners are bound via JavaScript.

Users navigating exclusively with a keyboard (or switch devices) **cannot focus on or activate any navigation item**, making the entire application inaccessible beyond the default home dashboard.

Additionally, all SVG interactive hotspots in the Diagram Studio are `<g>` groups with inline `onclick` handlers — they cannot be tabbed to or triggered via Enter/Space.

CRITICAL

WCAG 1.4.3: Color Contrast Failures

Multiple text elements fail the minimum 4.5:1 contrast ratio required for WCAG AA compliance:

ELEMENT	MODE	FOREGROUND	BACKGROUND	RATIO	REQUIRED
`--text-muted`	Light	#94a3b8	#ffffff	2.5:1	4.5:1 ❌
`--text-muted`	Dark	#64748b	#0f111f	2.8:1	4.5:1 ❌
`--primary`	Light	#6366f1	#ffffff	4.0:1	4.5:1 ❌

Subtitles, progress metrics, and description text are unreadable for users with moderate low vision.

LOW

WCAG 1.1.1: Missing Alt Text

The student avatar image in the Scholar Profile card (`index.html` line 1514) lacks a descriptive `alt` attribute. Screen readers will announce the raw Unsplash image URL instead.

SECTION 07

Interaction Design & Micro-animations

HIGH

Missing Touch Events on Mobile Canvas

The interactive canvas roadmaps in the Knowledge Engine handle `mousedown` , `mousemove` , and `mouseup` events, but lack equivalent touch event listeners (`touchstart` , `touchmove` , `touchend`).

The click-and-drag physics for roadmap nodes is completely **non-functional on mobile devices and tablets**.

MEDIUM

Native Blocking Alert Dialogs

Copying mnemonic notes triggers a native browser `alert()` modal. This breaks the premium glassmorphic visual aesthetic, pauses the entire rendering thread, and forces a jarring modal interaction inconsistent with the platform's design language.

RECOMMENDATION

Replace all `alert()` calls with a custom non-blocking toast notification component styled with the platform's glassmorphic design tokens.

SECTION 08

Competitive Benchmarking

Issac is benchmarked against leading medical learning platforms to identify competitive gaps and differentiation opportunities.

FEATURE	AMBOSS	OSMOSIS	ISSAC (CURRENT)	ISSAC (PROPOSED)
Visual Aesthetics	Standard	Cartoon/Flat	**Glassmorphism ✦**	**Glassmorphism ✦**
Simulation Realism	**High (Adaptive)**	Low (Linear MCQs)	**Broken (Cheat sheet)**	**High (Locked cards)**
Keyboard a11y	Compliant ✓	Compliant ✓	**Non-functional ✕**	Compliant ✓
Theme Flexibility	Light/Dark	Light/Dark	**Broken hover bug**	Flawless ✓
Mobile Touch	Dedicated Apps	Responsive Web	**Broken canvas**	Touch gestures ✓
Content Depth	**40K+ Articles**	Video-first	9 Topics + Generator	Scalable DB

Competitive Advantage

Issac's **glassmorphic design language** and **interactive SVG anatomy diagrams** are significant visual differentiators that no competitor currently offers. With the simulator learning loop restored and accessibility fixed, Issac can claim best-in-class simulation and design.

SECTION 09

Prioritized Recommendations & Strategic Roadmap

Phase 1: Quick Wins — Immediate Deployment

High-impact CSS and HTML fixes that require no architectural changes. Deployable within 1–2 days.

1. Fix Light Mode Button Hover Contrast

Add `body:not(.dark-mode) .btn-outline:hover` override to set `color: var(--primary)` and `border-color: var(--primary)` instead of white-on-white.

2. Remove Duplicate Selector in Concept Compare

Delete `#compare-select-2` and the "with" label from `index.html`. Relabel selector to "Select Comparison Pair:". Matches the actual JS data model.

3. Add Bookmark Button Active Styling

Define `.bookmark-btn.active` in CSS with `background: var(--primary-glow); color: var(--primary); border-color: var(--primary)` to provide immediate visual feedback.

4. Fix Text Muted Contrast Ratios

Adjust `--text-muted`: Light mode → `#64748b` (4.6:1 ratio). Dark mode → `#94a3b8` (4.8:1 ratio). Achieves WCAG AA compliance.

Phase 2: Core Usability Improvements — Sprint 1–2

Structural layout and logic changes that restore educational value and basic accessibility.

1. Redesign Case Simulator Layout Cockpit

Split the screen: Chat Simulator on the left, Patient Case File cards on the right. Move static reference cards into collapsible tabs to balance column heights.

2. Restore Locked/Blurred Card Progression

Re-enable lock overlays for the patient file stack. Start with only the Patient Summary visible. Progressively unlock Differentials → Investigations → Management as the user correctly answers each simulation step.

3. Implement Keyboard Accessibility

Add `tabindex="0"` and `role="button"` to all `.sidebar-item` elements and SVG `.svg-hotspot` groups. Bind keydown listeners to trigger click actions on Enter/Space.

Phase 3: Strategic UX Enhancements — Roadmap

Architectural improvements for scalability, mobile excellence, and design system maturity.

1. Consolidate Navigation Architecture

Unify 14 sidebar items into 5 primary workspace hubs: Dashboard, Learning Engine, AI Workspace, Simulation Labs, Creator Studio. Nest contextual widgets within parent hubs.

2. Implement Mobile Touch Support

Map `touchstart`, `touchmove`, and `touchend` events in the Knowledge Engine canvas to support node dragging on tablets and mobile devices.

3. Replace Native Alerts with Toast Notifications

Implement a non-blocking, glassmorphic toast notification component for "Copied to clipboard!", "Topic Bookmarked!", and error feedback messages.

4. Eliminate Inline Style Design Debt

Migrate all inline style overrides into the CSS design system. Create semantic badge variants (`.badge-primary`, `.badge-accent`) to replace ad-hoc inline color overrides.

SECTION 10

Implementation Checklist

A comprehensive checklist for designers, developers, and product stakeholders to track remediation progress.

Phase 1 — Quick Wins

- [ ] CSS override for `.btn-outline:hover` to fix white-on-white text bug in Light Mode
- [ ] Remove duplicate select box `#compare-select-2` from the Concept Comparison panel
- [ ] Add active visual state for `.bookmark-btn.active` to provide bookmarking feedback
- [ ] Adjust `--text-muted` color values for WCAG AA contrast compliance

Phase 2 — Core Usability

- [ ] Redesign Case Simulator layout: Chat on left, Patient File on right
- [ ] Restore locked/blurred card progression in the Clinical Case Simulator
- [ ] Add `tabindex="0"` and keyboard listeners to sidebar navigation items
- [ ] Add `tabindex="0"` and keyboard listeners to SVG interactive hotspots
- [ ] Remove legacy `panel-concept-maps` dead-end panel

Phase 3 — Strategic Enhancements

- [ ] Consolidate 14 sidebar items into 5 navigation hubs
- [ ] Add touch gesture support (`touchstart` / `touchmove`) to Knowledge Engine canvas
- [ ] Build and integrate glassmorphic toast notification component
- [ ] Migrate all inline style overrides into CSS design system classes

- [ ] Add descriptive `alt` attributes to all avatar and decorative images

End of Audit Report

Issac Healthcare AI Platform — UI/UX Audit & Strategic Roadmap

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